

MOHD. MUSHEER

Machine Learning Engineer | ML Systems & HPC

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in high-performance ML systems and scalable model architecture. Experienced in building ML algorithms from first principles using C++ and R, integrating BLAS/LAPACK for optimized and hardware-accelerated linear algebra, and implementing parallel processing with OpenMP. Strong foundation in mathematical optimization, system design, and production deployment using FastAPI and Docker. Dedicated to developing research-grade ML infrastructure grounded in mathematical rigor and performance optimization.

EDUCATION

Bachelor of Computer Applications (BCA)

Takshashila Mahavidyalaya, Amravati

Expected Graduation: May 2026

TECHNICAL SKILLS

Programming: **Python, SQL, R, C++**

Machine Learning: **Linear Models, Tree-Based Models, Ensemble Methods, Optimization Algorithms, Model Evaluation**

High-Performance Computing: **BLAS, LAPACK, OpenMP, Memory Optimization, Zero-Copy Data Exchange**

Deep Learning & NLP: PyTorch, Transformer Architectures (PRIMERA/LED), Hugging Face

Natural Language Processing: **TF-IDF, Cosine Similarity, Text Preprocessing**

Deployment & DevOps: FastAPI, Docker, REST APIs, CI/CD, Cloud Fundamentals (AWS/GCP)

Tools: **Git, GitHub, VS Code, Jupyter Notebook**

EXPERIENCE

Machine Learning Intern – Future Interns | Jan 2026 – Feb 2026 [Certificate](#)

- Developed NLP-based resume ranking system processing 1,000+ resumes using TF-IDF
- Performed EDA and feature engineering to improve model performance by 10%
- Evaluated classification models using Accuracy, Precision, Recall, and F1-score
- Awarded Letter of Recommendation for exemplary internship performance

PROJECTS

VectorForgeML — High-Performance ML Framework in R & C++ ([GitHub](#))

- Built a research-grade ML framework integrating R with a high-performance C++ backend
- Implemented core algorithms (Linear/Logistic Regression, Decision Trees, Random Forest, KNN, PCA, K-Means) from scratch
- Integrated BLAS/LAPACK for optimized linear algebra and OpenMP for parallel processing
- Designed modular Pipeline system with zero-copy R-C++ memory interface
- Published and accepted on **CRAN (Comprehensive R Archive Network)**

Indian News Summarizer (ROUGE-1: 71%) ([GitHub](#))

- Fine-tuned PRIMERA transformer model for abstractive summarization of Indian news articles
- Achieved ROUGE-1: 71% and BERTScore: 0.93 for factual consistency
- Built FastAPI backend with real-time inference and Dockerized for scalable deployment.

AI Resume Ranker & Candidate Screening System ([GitHub](#)) ([Live](#))

- Built ATS-style NLP system using TF-IDF and cosine similarity
- Implemented dynamic skill matching and candidate ranking
- Deployed using FastAPI and Docker